

RAK17000 WisBlock Motor Control Module Datasheet

Overview

Description

The RAK17000 is a DC motor driver module. It can be mounted to the IO Slot of the WisBlock Base of your choice. It can drive one stepper motor or two DC motors. The RAK17000 module uses the DRV8833 motor driver IC from Texas Instruments.

Features

- Drive two DC motors or one stepper motor
- Wide power supply voltage range: 2.7 to 10.8 V
- Motor power supply can be from an external source or from the WisBlock Base battery
- Output current control
- Selection of power source can be done via 2.54 mm 2-pin jumper
- Chipset: Texas Instruments DRV8833
- Module size: 25X35 mm

Applications

- Battery-Powered Mechanical Devices
- Automation Machines
- Robotics

Specifications

Overview

Mounting

The RAK17000 module can be mounted to the slots IO of the WisBlock Base board. Figure 1 shows the mounting mechanism of the RAK17000 on a WisBlock Base module, such as the RAK5005-O.

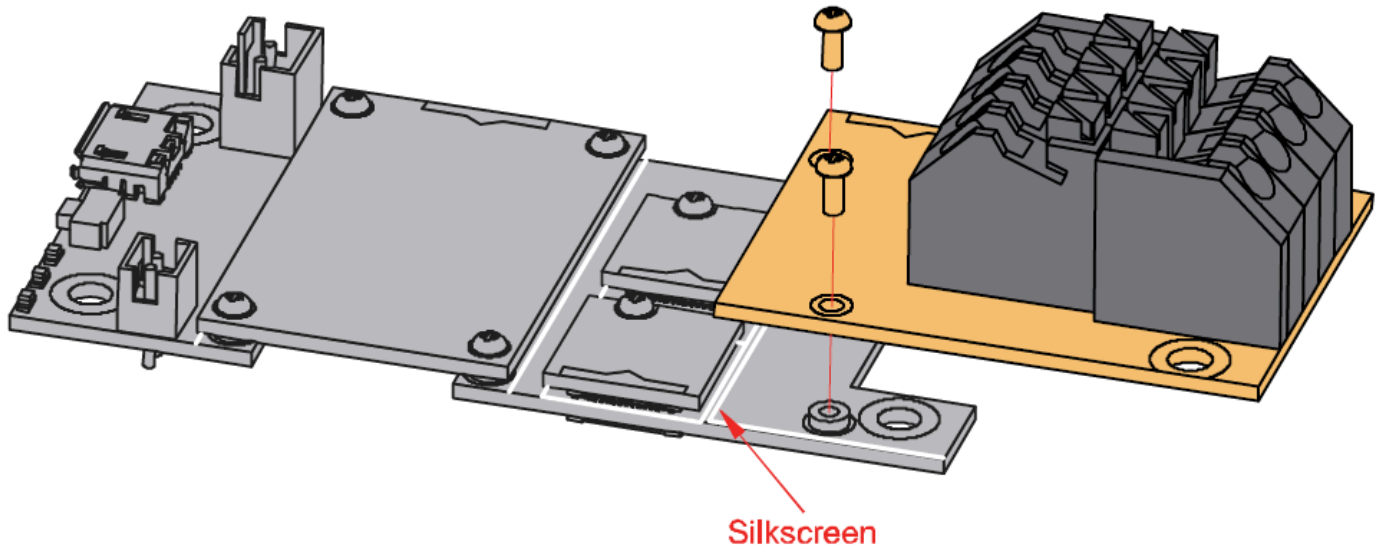


Figure 1: RAK17000 WisBlock Motor Drive Module Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts and the corresponding functions and diagrams of the module. It also covers the electrical and mechanical characteristics that include the tabular data of the functionalities and standard values of the RAK17000 WisBlock Motor Drive Module.

Chipset

The RAK17000 motor drive module uses DRV8833 from Texas Instruments.

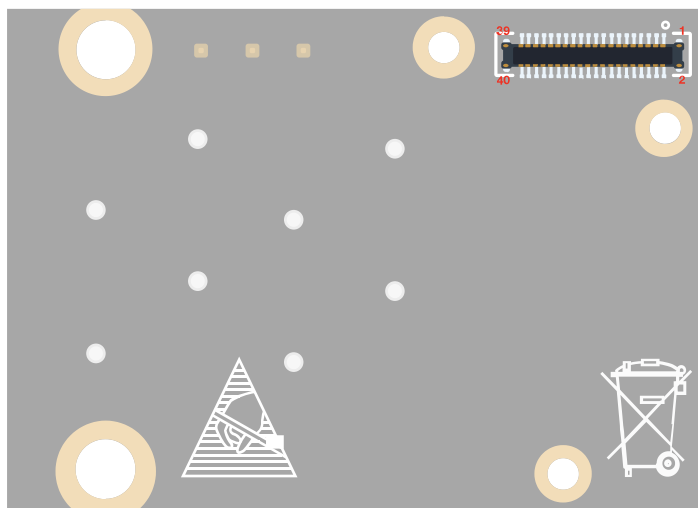
Vendor	Part Number
Texas Instruments	DRV8833

Pin Definition

The RAK17000 WisBlock motor drive module comprises a standard WisBlock connector for the IO slot. The WisBlock connector allows the RAK17000 module to be mounted on a WisBlock Base Board, such as RAK5005-O. The pin order of the connector and the pinout definition is shown in Figure 2.

NOTE

- Bridge **A** and **B** input related pins, **nSLEEP**, **VBAT**, and **GND** are connected to this connector.
- **nSLEEP** enable pin, high active, internal pull-down, low-power sleep mode



VBAT	2	1	VBAT
GND	4	3	GND
NC	6	5	NC
NC	8	7	NC
NC	10	9	NC
NC	12	11	NC
NC	14	13	NC
NC	16	15	NC
NC	18	17	NC
NC	20	19	NC
NC	22	21	NC
NC	24	23	NC
AIN1	26	25	AIN2
BIN2	28	27	NC
NC	30	29	NC
NC	32	31	BIN1
NC	34	33	NC
NC	36	35	NC
nSLEEP	38	37	NC
GND	40	39	GND

Figure 2: RAK17000 WisBlock Motor Drive Module Pinout

Electrical Characteristics

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VM	Device power supply (Motor and Driver IC)	2.7	-	10.8	V
VDIGIN	Digital input pin voltage range	-0.3	-	5.75	V
IOUT	Output current per bridge	-	-	0.85	A
IVM	VM sleep mode supply current	-	1.6	2.5	uA

Mechanical Characteristics

Board Dimensions

Figure 3 shows the dimensions and the mechanic drawing of the RAK17000 module.

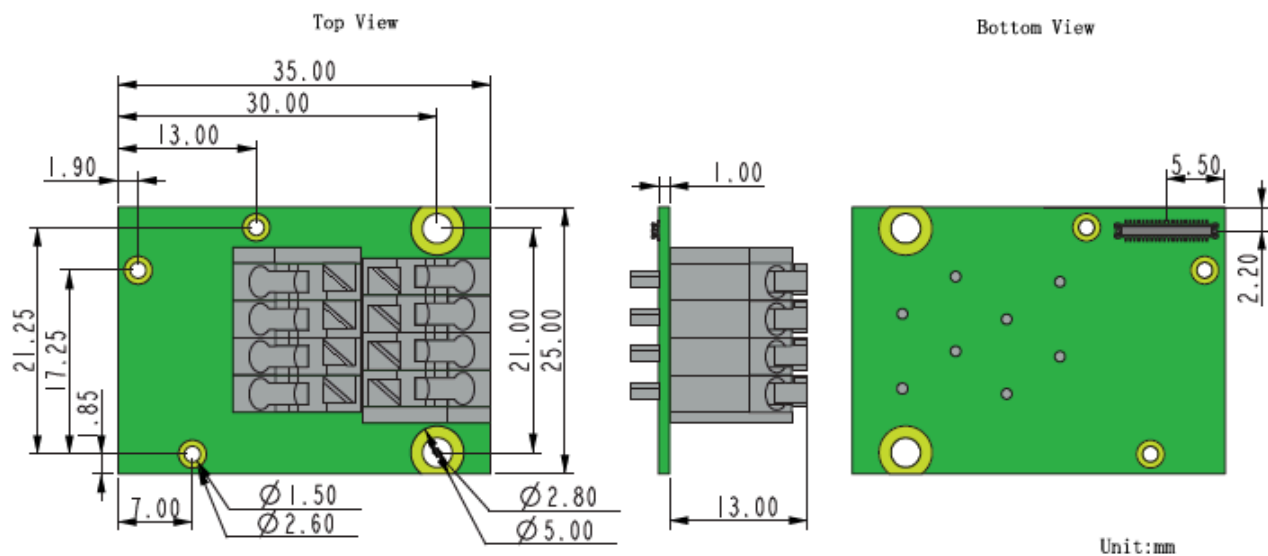


Figure 3: RAK17000 WisBlock Motor Control Module Mechanical Drawing

WisConnector PCB Layout

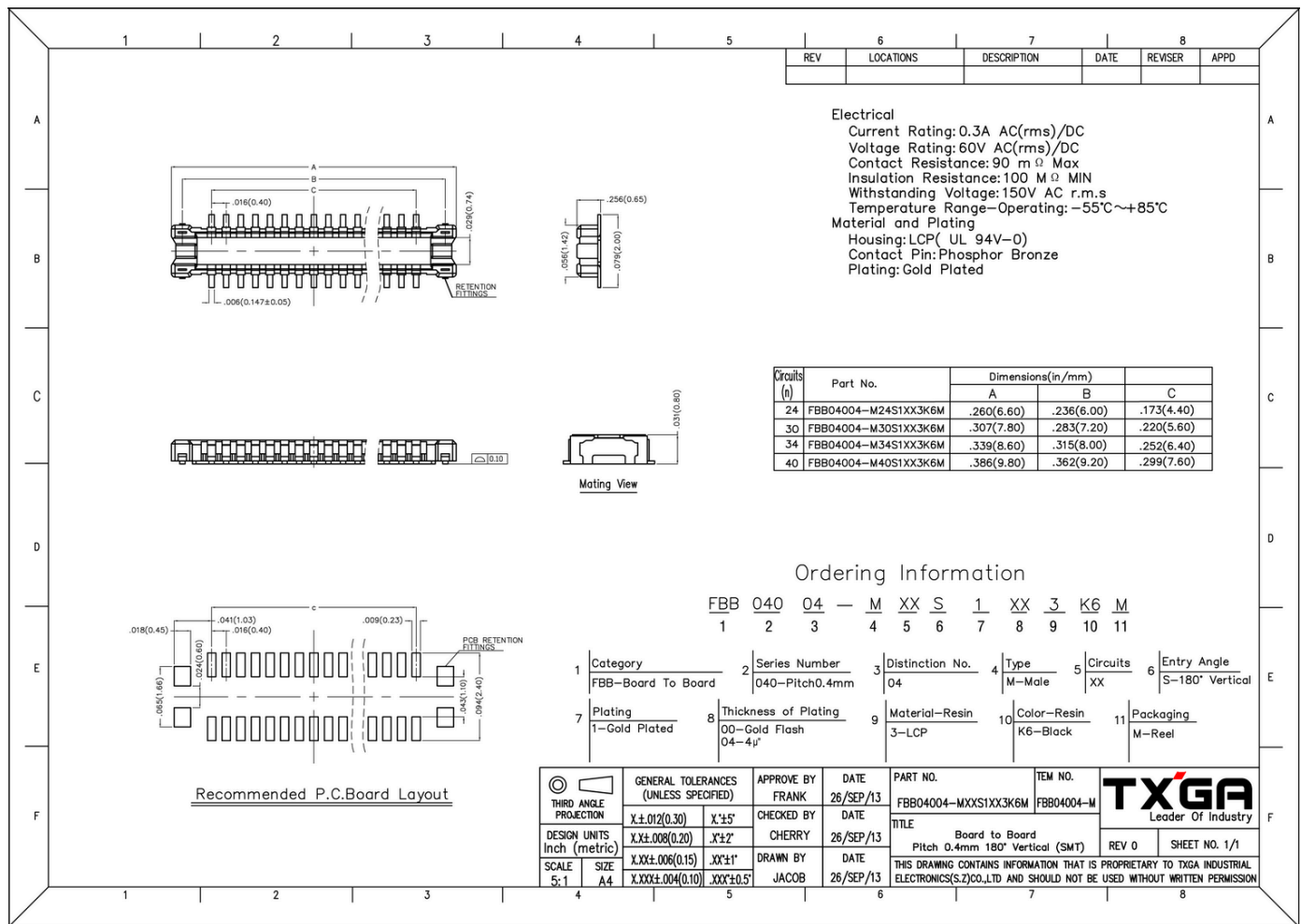


Figure 4: WisConnector PCB Footprint and Recommendations

Schematic Diagram

Motor Drive

Figure 5 shows RAK17000 motor drive schematic. VM is the motor power supply that has a range from 2.7 V to 10.8 V. It can change the motor current through modifying the R4, R5, R8, or R9. Default max output current is 0.85 A.

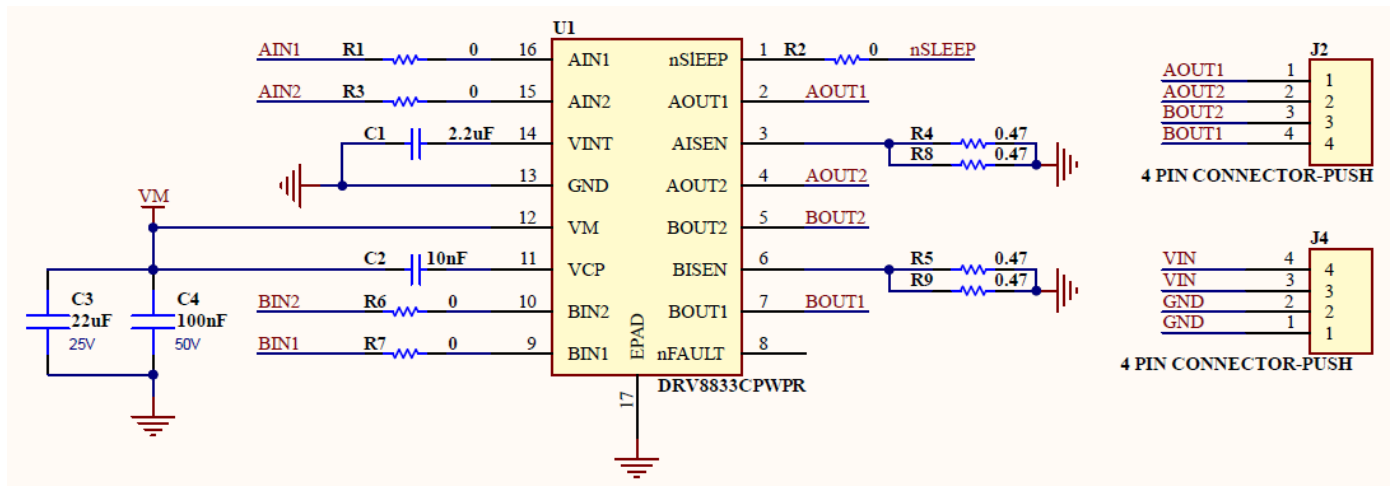


Figure 5: RAK17000 WisBlock Motor Drive schematic

Power Supply Switch

The power source of the motor can be selected via the 2.54 mm pitch jumper as shown in Figure 6. It can be from the battery of the WisBlock Base or via the external input voltage in the RAK17000 terminal. This is also the power source of the Driver IC. The max voltage input is 10.8 V.

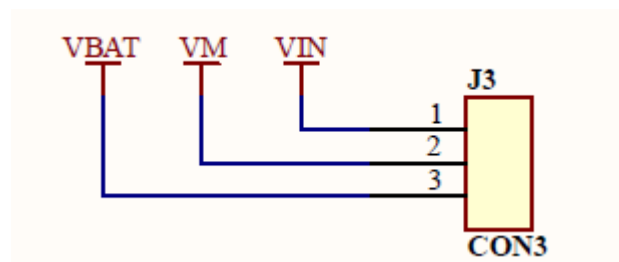


Figure 6: RAK17000 Wisblock Power Supply Switch